BEFORE YOU READ

Inverses of the reciprocal functions

Each inverse reciprocal function can be rewritten with an ordinary inverse function of $1/x$. Their domains and ranges come with gaps.

Inverse reciprocal functions

arccsc · arcsec · arccot

arccsc	arcsec	arccot
$\arcsin(1/x)$	$\arccos(1/x)$	$\arctan(1/x)$ for $x > 0$
Domain $\lvert x\rvert \geq 1$	Domain $\lvert x\rvert \geq 1$	Domain all reals
Range $[-\pi/2, \pi/2]$ without 0	Range $[0, \pi]$ without $\pi/2$	Range $(0, \pi)$

TAKEAWAY

Reduce to arcsin, arccos, arctan

For computation, switch to $\arcsin(1/x)$, $\arccos(1/x)$ or $\arctan(1/x)$, watching the domain each one needs.